

Hierarchically Distributed Energy Management in Distribution Systems: An Error-Tolerant and Asynchronous Approach

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Abstract—Distribution systems are becoming more complex with the integration of numerous distributed energy resources (DERs), such as distributed generators (DGs), distributed energy storages (DESs), and controllable end-users (EUs). Aggregators (Aggs) are emerging to bridge the gap between the distribution system operator (DSO) and EUs. This creates a hierarchical “DSO-Agg-EU” architecture that requires effective and efficient coordination among massive agents. However, this architecture faces challenges such as communication asynchrony and solution inexactness among different agents, which may undermine the performance of energy management or even cause unexpected failures. To address these challenges, this paper proposes an error-tolerant and asynchronous algorithm based on dual decomposition for hierarchically distributed energy management (HDEM). The algorithm is theoretically guaranteed to converge to a near-optimal value and can be further accelerated by Nesterov’s method. The algorithm is tested on the IEEE 123-bus distribution system and demonstrates high effectiveness and efficiency under asynchrony and inexactness.

Index Terms—Energy management, hierarchical architecture, distributed optimization, dual decomposition, asynchronous algorithm, inexact algorithm.

NOMENCLATURE

Abbreviations

ADMM	Alternating direction method of multipliers.
Agg	Aggregator.
ALM	Augmented Lagrangian method.
CL	Curtailable load.
DL	Deferrable load.
DER	Distributed energy resource.
DES	Distributed energy storage.
DG	Distributed generator.
DSO	Distribution system operator.

EU	End-user.
HDEM	Hierarchically distributed energy management.
PCC	Point of common coupling.
LinDistFlow	Linearized distribution flow.

Sets

$\mathcal{N}$	Set of bus.
$\mathcal{L}$	Set of distribution lines.
$\mathcal{L}_i^{eu}$	Set of distribution lines in the local community at bus i .
$\mathcal{U}_i$	Set of EUs at bus i .
$\mathcal{K}_i, \mathcal{K}_D$	Sets of iterative indexes of DER i and DSO.
$\mathcal{H}_j, \mathcal{H}_D$	Sets of iterative indexes of EU j and Agg.
$\mathcal{P}_i, \mathcal{P}_j^a$	Feasible regions of DER i and EU j .

Variables

$P_{ij,t}, Q_{ij,t}$	Active and reactive power flow from bus i to j at time t .
$l_{ij,t}$	Squared magnitude of current from bus i to j at time t .
$v_{i,t}$	Voltage of bus i at time t .
$p_{i,t}$	Injected power of DER i at time t .
$p_{i,t}^{ch}, p_{i,t}^{dc}$	Charging and discharging power of DES i at time t .
$e_{i,t}$	Stored energy of DES i until time t .
$p_{j,t}^{eu}$	Load of EU j at time t .
$p_{j,t}^{cl}, p_{j,t}^{dl}$	Curtailable and deferrable load of EU j at t .
$e_{j,t}^{dl}$	Provided energy of deferrable load of EU j until time t .
λ, ξ	Lagrangian multipliers.
μ, ζ	Auxiliary variables.

Parameters and Constants

T	Length of time slots.
Δt	Duration of time slot.
$p_{i,t}^d, q_{i,t}^d$	Active and reactive load at bus i at time t .
r_{ij}, x_{ij}	Resistance and reactance of line (i, j) .
$\underline{v}, \bar{v}$	Lower and upper bounds of bus voltage.
a_i^{DG}, b_i^{DG}	Cost parameters of DG i .
$\underline{P}_i^{DG}, \bar{P}_i^{DG}$	Lower and upper bounds of power output of DG i .
R_i	Maximal ramping rate of DG i .
a_i^{ch}, a_i^{dc}	Cost parameters of DES i .

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$\bar{P}_i^{DES}$	Maximal charge-discharge power of DES i .
$E_i, \bar{E}_i$	Lower and upper bounds of stored energy of DES i .
η_i^{ch}, η_i^{dc}	Charging and discharging efficiency of DES i .
$a_i^{agg}, P_{i,t}^{agg}$	Cost parameters of Agg i .
$\underline{P}_{i,t}^{agg}, \bar{P}_{i,t}^{agg}$	Lower and upper bounds of power output of Agg i .
F_l	Capacity of distribution line l in local community.
π_{jl}	Power flow distribution factor from EU j to distribution line l .
a_j^{EU}	Cost parameter of EU j .
$P_{j,t}^{EU}$	Expected consumption of EU j at time t .
$\underline{P}_{j,t}^{CL}, \bar{P}_{j,t}^{CL}$	Lower and upper bounds of curtailable load of EU j .
$\bar{P}_j^{DL}$	Upper bounds of deferrable load of EU j .
t_j^s, t_j^e	Start and termination time of energy requirement of EU j .
$\underline{E}_j^{DL}, \bar{E}_j^{DL}$	Lower and upper bounds of provided energy of deferrable load of EU j .
E_j^{tgt}	Targeted energy requirement of deferrable load of EU j .
α, α^a	Step size.
β_k, β_h^a	Nesterov's acceleration rate.
k_0, h_0	Communication asynchrony parameters.
$\varepsilon_i, \varepsilon_j$	Solution Inexactness parameters.

I. INTRODUCTION

THE PROLIFERATION of distributed energy resources (DERs), including but not limited to distributed generators (DGs), distributed energy storages (DESs), and controllable end-users (EUs), offers both opportunities and challenges for managing distribution systems. These DERs can facilitate renewable energy accommodation [1], voltage regulation [2], frequency control [3], and valley filling [4], which remarkably enhances the reliability while reducing the operational cost of the distribution system [5]. Moreover, when numerous small-capacity controllable EUs, such as electric vehicles (EVs), air conditioners, and other domestic appliances, are integrated into distribution networks, aggregators (Aggs) are commonly introduced as an intermediary between the distributed system operator (DSO) and EUs to relieve DSO's communicational and computational burden [6]. In this context, Aggs coordinate the power generation/consumption of their EUs, while DSO regards each Agg together with all its EUs as one DER to manage, rendering a hierarchical "DSO-Agg-EU" architecture of distributed networks (see Fig. 1). Even though, how to effectively manage massive DERs dispersed across the distribution network is still challenging.

The distribution system energy management has been widely investigated in the literature. Initially, DSO manages DERs in a centralized manner [6], [7]. DSO collects abundant information from all controllable agents, solves a global optimization problem, and then sends the optimal operation strategy back to agents, which highly challenges DSO's abilities of communication and computation.

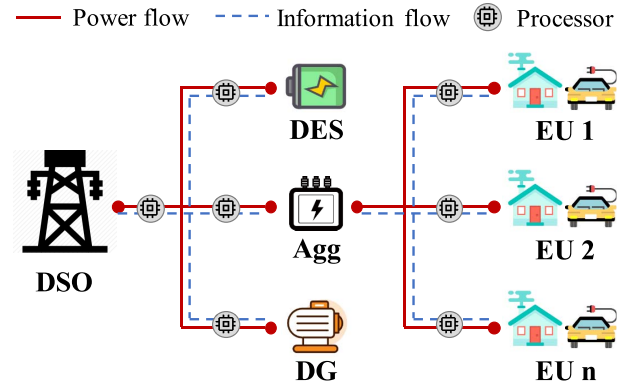


Fig. 1. Hierarchical DSO-Agg-EU architecture of distribution networks.

Besides, centralized energy management faces difficulty in preserving privacy since most of the DERs are personally owned [8].

To address these challenges, there has been an increasing interest in the distributed energy management of distribution systems [9] in recent years. From the interaction architecture, distributed energy management methods can be classified into three categories: hierarchical [4], fully distributed [3], and local [10]. In the hierarchical architecture, a coordinator will receive information of DERs and broadcast coordination signals. The hierarchical architecture can be stacked as a pyramid, which is known as the hierarchically distributed energy management (HDEM) architecture. Compared to single-layer distributed methods [3], [4], [10], HDEM approaches can manage massive EUs in large-scale distribution systems. Benefiting from the hierarchical "DSO-Agg-EU" architecture (see Fig. 1), the HDEM approaches can further alleviate the communicational and computational burdens compared to single-layer distributed methods. In [11], a decentralized robust model of smart grid operator coordinated with Aggs was built to combat uncertain wind generations and wholesale market prices. Reference [12] proposed a hierarchical updating method for the primal-dual gradient algorithm according to the radial topology of the distribution network, which remarkably reduced the communicational burden. In [13], a two-tier microgrid power scheduling was formulated to efficiently manage power trading between a macrogrid and cooperative microgrids, where the online and distributed algorithms were proposed to solve problems. Reference [14] proposed a two-level energy management framework to reduce the computational burden by allocating subproblems to EUs. In [15], a hierarchically distributed EV charging scheduling method was proposed based on the auxiliary problem principle to deal with the coupled quadratic normalized item. Reference [16] developed an HDEM method in three-phase unbalanced distribution systems, which can coordinate with heterogeneous loads, e.g., curtailable, deferrable, and interruptible loads.

The above studies [11], [12], [13], [14], [15], [16] are applied under the assumption that the iterative steps can be accurately and punctually implemented. However, in practical systems, there are always imperfect issues during iterations, e.g., communication asynchrony and solution inexactness. For

instance, in Fig. 1, the information flow may face time delay or packet drop, while the solution by the smart processor may introduce errors.

Communication asynchrony may originate from information loss, time delay in communication links, or different speeds of solving subproblems, etc. Communication asynchrony has a significant impact on the convergence speed of distributed energy management, which has attracted much attention [2], [4], [17], [18], [19]. Reference [4] proposed an asynchronous and distributed EV charging scheduling method, which can handle the time delay between the utility company and EVs. In [17], an asynchronous primal-dual gradient method was applied for the distributed optimal power flow. In [18] and [2], asynchronous distributed methods were proposed to solve the economic dispatch problem of microgrids and the voltage control problem with DERs, respectively. Reference [19] extended the distributed energy management problem to a multi-energy one, where the asynchronous iterations based on the dual decomposition method were considered.

Solution inexactness may be due to the preset error tolerances of the solver and the accuracy of parameters when agents solve subproblems. Solution inexactness may lead to economic damages or even operational accidents, which has been studied in [20], [21], [22], [23], [24]. Reference [20] proposed a distributed and real-time economic dispatch based on the alternating direction method of multipliers, where inexact solutions were acceptable with restriction to the computing power. In [21], a distributed optimal power flow of microgrids was developed, where a penalty-based optimization was utilized to reduce the error from the inexact second-order cone relaxation. Reference [22] showed the convergences of the dual decomposition method and its accelerated version in $\mathcal{O}(1/k)$ and $\mathcal{O}(1/k^2)$ when subproblem solutions were inexact. Besides, [23] proved the convergence of the dual decomposition method under the condition of the inexact gradient. Reference [24] coped with the lost metering messages in the distributed residential load control.

However, there is still a large gap. None of the above literature simultaneously considers communication asynchrony and solution inexactness in the distributed energy management problem, although they always co-exist in practical systems. In this paper, according to the hierarchical “DSO-Agg-EU” architecture, an error-tolerant and asynchronous HDEM (ETA-HDEM) algorithm is devised leveraging dual decomposition to efficiently solve the energy management problem. The proposed algorithm owns the following salient features:

- 1) *Robustness to asynchrony and computation errors.* Existing literature only consider either communication asynchrony [2], [4], [17], [18], [19] or solution inexactness [20], [21], [22], [23], [24] separately, while they always co-exist in practical systems. Compared to the existing works, the proposed algorithm tackles these two issues simultaneously, enabling an ETA-HDEM.
- 2) *Guarantees of convergence and performance.* The optimality and feasibility of the energy management strategy generated by the ETA-HDEM algorithm are theoretically guaranteed under communication asynchrony and solution inexactness. It is proved that the total operational

cost converges to a near-optimal value with an upper bound, and the violation of physical constraints diminishes to zero.

- 3) *Fast convergence rate.* By embedding a Nesterov’s acceleration into the proposed algorithm, the algorithm exhibits a fast convergence rate, which appears to be more applicable in large-scale distribution networks.

The rest of this paper is organized as follows. In Section II, the multi-period energy management problem of the distribution system with massive EUs is established. Section III proposes a primitive HDEM algorithm based on dual decomposition, while Section IV improves the algorithm to jointly tackle communication asynchrony and solution inexactness and proves its convergence. In Section V, the performance of the proposed algorithm is verified by numerical experiments. Section VI concludes the paper.

Notation: Let $\otimes$ denote the Kronecker product. For a sequence of scalars $\{x_{i,t}\}$ indexed by $i = 1, 2, \dots, N$ and $t = 1, 2, \dots, T$, define the compact form as

$$\mathbf{x} = (\mathbf{x}_1^T, \mathbf{x}_2^T, \dots, \mathbf{x}_N^T)^T \quad (1a)$$

$$\mathbf{x}_i = (x_{i,1}, x_{i,2}, \dots, x_{i,T})^T, \quad i = 1, 2, \dots, N. \quad (1b)$$

Denote by $\mathbf{1}_N$ and I_N the N -dimensional vector of all ones and the $N \times N$ identity matrix, respectively. For a set $\mathcal{K}$, denote its cardinality by $|\mathcal{K}|$. Define $[\mathbf{x}]^+ := \max\{\mathbf{x}, 0\}$. The norms of vectors and matrices refer to the 2-norm in this paper.

II. MATHEMATICAL FORMULATION

We focus on the energy management problem of a distribution system with a hierarchical “DSO-Agg-EU” architecture, which is shown in Fig. 1. The target of the energy management problem is to decide the operation/consumption strategies of DERs¹ and EUs to be implemented. The problem can be formulated as an optimization problem, in which the total operational costs of DERs and EUs are minimized with restrictions to operational constraints of DERs and EUs and network constraints, e.g., power balance, voltage limitations, and distribution line capacity limitations. In this paper, we focus on the decision phase to make the operation/consumption strategies of DERs and EUs. The automation and regulation dynamics in the control phase are not considered in the formulated energy management problem.

A. Power Flow Model

Consider a radial distribution network defined by the bus set $\mathcal{N}^+ = \{0\} \cup \mathcal{N}$ and the line set $\mathcal{L}$. Bus 0 is the point of common coupling (PCC) with a fixed voltage, while $\mathcal{N} = \{1, 2, \dots, N\}$ is the set of power injection buses. Let $\mathcal{L}_i \subset \mathcal{L}$ denote the line set of the unique path from bus i to the PCC. In the manuscript, we consider a multi-period energy management problem with T time slots.

¹For simplification, Aggs, DGs, and DESs are uniformly referred to as DERs hereafter.

From [25], the distribution flow (DistFlow) model is given for each line $(i, j) \in \mathcal{L}$, $t = 1, 2, \dots, T$ as

$$P_{ij,t} = p_{j,t}^d - p_{j,t} + r_{ij}l_{ij,t} + \sum_{(j,k) \in \mathcal{L}} P_{jk,t} \quad (2a)$$

$$Q_{ij,t} = q_{j,t}^d + x_{ij}l_{ij,t} + \sum_{(j,k) \in \mathcal{L}} Q_{jk,t} \quad (2b)$$

$$v_{i,t}^2 - v_{j,t}^2 = 2(r_{ij}P_{ij,t} + x_{ij}Q_{ij,t}) - (r_{ij}^2 + x_{ij}^2)l_{ij,t} \quad (2c)$$

$$l_{ij,t}v_{i,t}^2 = P_{ij,t}^2 + Q_{ij,t}^2. \quad (2d)$$

By [2], [10], [26], the net loss is much smaller than the line power, i.e., $r_{ij}l_{ij,t} \ll P_{ij,t}$, $x_{ij}l_{ij,t} \ll Q_{ij,t}$. Reference [10] shows that the relative error introduced by ignoring the net loss is only at the order of 1%. By ignoring the network loss and defining $V_{i,t} = v_{i,t}^2/2$ [2], the linearized distribution flow (LinDistFlow) model is given for each line $(i, j) \in \mathcal{L}$, $t = 1, 2, \dots, T$ as

$$P_{ij,t} = p_{j,t}^d - p_{j,t} + \sum_{(j,k) \in \mathcal{L}} P_{jk,t} \quad (3a)$$

$$Q_{ij,t} = q_{j,t}^d + \sum_{(j,k) \in \mathcal{L}} Q_{jk,t} \quad (3b)$$

$$V_{i,t} - V_{j,t} = r_{ij}P_{ij,t} + x_{ij}Q_{ij,t}. \quad (3c)$$

Suppose the voltage of PCC is fixed and time-invariant, i.e., $V_0 := v_{0,t}^2/2, \forall t$. We have the compact form of the LinDistFlow model as

$$\begin{aligned} \mathbf{V} &= V_0 \mathbf{1}_{NT} - R(\mathbf{p}^d - \mathbf{p}) - X\mathbf{q}^d \\ &= \mathbf{V}_B + R\mathbf{p}, \end{aligned} \quad (4)$$

where $\mathbf{V}_B = V_0 \mathbf{1}_{NT} - R\mathbf{p}^d - X\mathbf{q}^d$ is the baseline of the voltage profile, and vectors $\mathbf{p}, \mathbf{p}^d, \mathbf{q}^d$, and $\mathbf{V}$ are compact forms of sequences $\{p_{i,t}\}, \{p_{i,t}^d\}, \{q_{i,t}^d\}$, and $\{V_{i,t}\}$, respectively. Matrices R and X are defined as

$$R = \check{R} \otimes I_T, \quad X = \check{X} \otimes I_T, \quad (5)$$

where elements of matrices $\check{R}$ and $\check{X}$ are given by

$$\check{R}_{ij} = \sum_{(i',j') \in \mathcal{L}_i \cap \mathcal{L}_j} r_{i'j'}, \quad \check{X}_{ij} = \sum_{(i',j') \in \mathcal{L}_i \cap \mathcal{L}_j} x_{i'j'}. \quad (6)$$

The bus voltage constraint is given by

$$\frac{v^2}{2} \mathbf{1}_{NT} \leq \mathbf{V}_B + R\mathbf{p} \leq \frac{\bar{v}^2}{2} \mathbf{1}_{NT}. \quad (7)$$

B. Models of DERs and EUs

There are mainly three types of DERs considered in the distribution system energy management problem, e.g., DGs, DESs, and Aggs. Note that $p_{i,t}$ is the unified power output from DER i to the distribution network at time slot t . The Agg at bus i coordinates with a set of EUs in a local community denoted by $\mathcal{U}_i$.

For an EU, there are various residential appliances in the house, which mainly fall into two categories: curtailable load (CL) and deferrable load (DL). For CLs, e.g., air conditioners and electric heaters, the operation power can be adjusted within a given time interval. For DLs, e.g., electric vehicles

(EVs) and washing machines, a given energy requirement should be satisfied in certain time slots, at which the power can be flexibly allocated.

1) *Distributed Generator*: For the DG at bus i , its generation cost is a quadratic function

$$C_i^{DG} = \sum_{t=1}^T a_i^{DG} p_{i,t}^2 + b_i^{DG} p_{i,t}. \quad (8)$$

Operational constraints of DG i are formulated by

$$\underline{p}_i^{DG} \leq p_{i,t} \leq \bar{p}_i^{DG}, \quad t = 1, 2, \dots, T \quad (9a)$$

$$-R_i \leq p_{i,t+1} - p_{i,t} \leq R_i, \quad t = 1, 2, \dots, T-1, \quad (9b)$$

where (9a) and (9b) represent the generation power and ramping limitations, respectively.

2) *Distributed Energy Storage*: For the DES at bus i , from [27], the cost of battery degradation is given by

$$C_i^{DES} = \sum_{t=1}^T a_i^{ch} (p_{i,t}^{ch})^2 + a_i^{dc} (p_{i,t}^{dc})^2. \quad (10)$$

Operational constraints of DES i are formulated by

$$p_{i,t} = p_{i,t}^{dc} - p_{i,t}^{ch}, \quad t = 1, 2, \dots, T \quad (11a)$$

$$0 \leq p_{i,t}^{dc} \leq \bar{p}_i^{DES}, \quad t = 1, 2, \dots, T \quad (11b)$$

$$0 \leq p_{i,t}^{ch} \leq \bar{p}_i^{DES}, \quad t = 1, 2, \dots, T \quad (11c)$$

$$\underline{e}_i \leq e_{i,t} \leq \bar{e}_i, \quad t = 1, 2, \dots, T \quad (11d)$$

$$e_{i,t} = e_{i,t-1} + (\eta_i^{ch} p_{i,t}^{ch} - p_{i,t}^{dc} / \eta_i^{dc}) \Delta t, \quad t = 1, 2, \dots, T \quad (11e)$$

$$e_{i,T} = e_{i,0}, \quad (11f)$$

where (11b)–(11d) are power and energy physical limitations, (11e) describes the process of battery discharging/charging, and (11f) indicates the energy balance during the whole period.

3) *Aggregator*: The Agg at bus i coordinates with a set of EUs in a local community to satisfy some intent with respect to the total load profile. The cost of Agg is formulated as

$$C_i^{Agg} = \sum_{t=1}^T a_i^{Agg} (p_{i,t} - P_{i,t}^{Agg})^2, \quad (12)$$

If $P_{i,t}^{Agg}$ is a given load profile, the objective of Agg can be tracking the expected load profile. When setting $P_{i,t}^{Agg} = 0, \forall t$, the objective is flattening the total load profile [4]. Operational constraints of Agg i include

$$\underline{p}_{i,t}^{Agg} \leq p_{i,t} \leq \bar{p}_{i,t}^{Agg}, \quad t = 1, 2, \dots, T \quad (13a)$$

$$p_{i,t} = - \sum_{j \in \mathcal{U}_i} p_{j,t}^{eu}, \quad t = 1, 2, \dots, T \quad (13b)$$

$$-F_l \leq \sum_{j \in \mathcal{U}_i} \pi_{jl} p_{j,t}^{eu} \leq F_l, \quad l \in \mathcal{L}_i^{eu}, t = 1, 2, \dots, T, \quad (13c)$$

where constraint (13a) is the physical limitation of power, equation (13b) shows that the total load profile of Agg i is the aggregation of EUs' loads, and inequality (13c) restricts loads based on distribution line capacities.

4) *End-User*: The EU $j \in \mathcal{U}_i$ owns a baseline of its expected power consumption $P_{j,t}^{EU}$, $\forall t$. When the load is changed, the dis-utility function is presented to describe the dissatisfaction of the EU, i.e.,

$$C_j^{EU} = \sum_{t=1}^T a_j^{EU} (p_{j,t}^{eu} - P_{j,t}^{EU})^2, \quad (14)$$

where a_j^{EU} is constant. Operational constraints of EU j are formulated by

$$p_{j,t}^{eu} = p_{j,t}^{cl} + p_{j,t}^{dl}, t = 1, 2, \dots, T \quad (15a)$$

$$\underline{p}_{j,t}^{CL} \leq p_{j,t}^{cl} \leq \bar{p}_{j,t}^{CL}, t = 1, 2, \dots, T \quad (15b)$$

$$0 \leq p_{j,t}^{dl} \leq \bar{p}_j^{DL}, t_j^s \leq t \leq t_j^e \quad (15c)$$

$$p_{j,t}^{dl} = 0, t < t_j^s \text{ or } t > t_j^e \quad (15d)$$

$$\underline{E}_j^{DL} \leq e_{j,t}^{dl} \leq \bar{E}_j^{DL}, t = 1, 2, \dots, T \quad (15e)$$

$$e_{j,t}^{dl} = e_{j,t-1}^{dl} + p_{j,t}^{dl} \Delta t, t = 1, 2, \dots, T \quad (15f)$$

$$e_{j,t}^{dl} \geq E_j^{tgt}, t = t_j^e, \quad (15g)$$

where (15b)–(15e) are physical limitations of power and energy, (15f) describes the process of the DL energy task, and (15g) identifies the final energy requirement.

C. Overall Energy Management Problem

The objective of the energy management problem is to minimize the total operational costs of all DERs and EUs subject to power flow constraints of the distribution system and operational constraints of DERs and EUs, which is defined as

$$\min \sum_i C_i^{DG} + \sum_i C_i^{DES} + \sum_i \left(C_i^{Agg} + \sum_{j \in \mathcal{U}_i} C_j^{EU} \right) \quad (16a)$$

$$\text{s.t. (7), (9), (11), (13), (15).} \quad (16b)$$

The rapidly increasing DERs and EUs cause high computational costs to solve the energy management problem (16). This paper proposes an HDEM algorithm based on dual decomposition, which is a two-loop distributed iterative algorithm. Section III establishes the primitive algorithm, while Section IV extends the algorithm to an error-tolerant and asynchronous version against both communication asynchrony and solution inexactness.

III. PRIMITIVE HDEM WITH DUAL DECOMPOSITION

In this section, we devise a primitive two-loop HDEM algorithm based on the dual decomposition method. The algorithm is established based on assumptions that communications among agents are ideal and agents always obtain exact solutions during iterations.

The whole distribution system can be regarded as a multi-agent system (MAS) with three types of agents, DG, DES, and Aggs (coordinating with EUs). The outer-loop iteration algorithm divides the whole energy management problem into subproblems of agents. The subproblem of Agg can be also regarded as an optimization problem in a MAS with

two types of agents, Agg, and EUs. Then the subproblem of Agg is divided by the inner-loop iteration. The basic distributed algorithm in the outer- and inner-loop iterations is dual decomposition.

A. Outer-Loop Iteration

The outer-loop iteration is the distributed optimization between DSO and DERs. Problem (16) can be simplified by

$$\min_{\mathbf{p}} \sum_{i \in \mathcal{N}} f_i(\mathbf{p}_i) \quad (17a)$$

$$\text{s.t. } \mathbf{p}_i \in \mathcal{P}_i, \forall i \in \mathcal{N} \quad (17b)$$

$$A\mathbf{p} \leq \mathbf{b}, : \lambda, \quad (17c)$$

where $f_i(\mathbf{p}_i)$ and $\mathcal{P}_i$ represent the operational cost and constraints of DER i defined by (8)–(15), respectively, constraint (17c) is the compact form of (7), and λ is the Lagrangian multiplier of (17c).

Assumption 1: 1) The operational cost $f_i(\mathbf{p}_i)$ of DER i is c_i -strongly convex and differentiable, $\forall i \in \mathcal{N}$.

2) The feasible region $\mathcal{P}_i$ of DER i is nonempty, compact, and convex, $\forall i \in \mathcal{N}$.

3) There exists a strictly feasible interior point in (17).

Since the objective function (17a) is quadratic and the constraints (17b) and (17c) are linear, Assumption 1 obviously holds.

Define the Lagrangian of (17)

$$L(\mathbf{p}; \underline{\lambda}, \bar{\lambda}) = \sum_{i \in \mathcal{N}} f_i(\mathbf{p}_i) + \lambda^T (A\mathbf{p} - \mathbf{b}) \quad (18)$$

and the auxiliary variable

$$\mu = A^T \lambda. \quad (19)$$

Based on dual decomposition, we have the following distributed algorithm indexed by k

$$\mathbf{p}_i(k) = \arg \min_{\mathbf{p}_i \in \mathcal{P}_i} f_i(\mathbf{p}_i) + \mu_i^T(k) \mathbf{p}_i \quad (20a)$$

$$\lambda(k+1) = [\lambda(k) + \alpha(A\mathbf{p}(k) - \mathbf{b})]^+ \quad (20b)$$

$$\mu(k+1) = A^T \lambda(k+1), \quad (20c)$$

where $\alpha > 0$ is the step size, (20a) and (20b)–(20c) are DER's step of solving the subproblem and DSO's step of updating Lagrangian multipliers, respectively. The auxiliary variable follows $\mu(k) := (\mu_1^T(k), \mu_2^T(k), \dots, \mu_N^T(k))^T$. During iterations, only $\mathbf{p}_i(k)$ and $\mu_i(k)$ are transferred between DSO and DER i .

DGs and DESs can directly and easily solve subproblem (20a). However, for Agg i , the subproblem is still difficult to solve due to the vast number of EUs. A distributed optimization method is required to solve Agg's subproblem.

B. Inner-Loop Iteration

At each iteration step k , the subproblem of Agg i owns the same mathematical structure as (17). Regard the Agg as a "special" EU and define $\mathcal{U}_i^+ = \{i\} \cup \mathcal{U}_i$. For brevity, omit the indexes k and i in the inner-loop iteration. Then the

subproblem of Agg (20a) is extended to

$$\min_{\mathbf{p}^a} \sum_{j \in \mathcal{U}_i^+} g_j(\mathbf{p}_j^a) \quad (21a)$$

$$\text{s.t. } \mathbf{p}_j^a \in \mathcal{P}_j^a, \forall j \in \mathcal{U}_i^+ \quad (21b)$$

$$C\mathbf{p}^a \leq \mathbf{d}, : \xi, \quad (21c)$$

where $\mathbf{p}_j^a$ represents EU's load $\mathbf{p}_j^{eu}$ or Agg's power output $\mathbf{p}_i$, $\mathbf{p}^a$ is the compact form of $\mathbf{p}_j^a, j \in \mathcal{U}_i^+$, $g_j(\mathbf{p}_j^a)$ is the cost of Agg, $C_i^{Agg} - \mu_i^T(k) \mathbf{p}_i^a$, or the cost of EU, C_j^{EU} , the feasible region $\mathcal{P}_j^a$ is defined by (13a) for Agg and (15) for EU, constraint (21c) is the compact form of (13b)–(13c), and ξ is the Lagrangian multiplier of (21c).

Define the Lagrangian of (21)

$$L^a(\mathbf{p}_j^a; \xi) = \sum_{j \in \mathcal{U}_i^+} g_j(\mathbf{p}_j^a) + \xi^T (C\mathbf{p}^a - \mathbf{d}). \quad (22)$$

Applying dual decomposition, we obtain the following distributed algorithm indexed by h

$$\mathbf{p}_j^a(h) = \arg \min_{\mathbf{p}_j^a \in \mathcal{P}_j^a} g_j(\mathbf{p}_j^a) + \xi_j^T(h) \mathbf{p}_j^a \quad (23a)$$

$$\xi(h+1) = [\xi(h) + \alpha^a (C\mathbf{p}_j^a(h) - \mathbf{d})]^+ \quad (23b)$$

$$\zeta(h+1) = C^T \xi(h+1), \quad (23c)$$

where $\alpha^a > 0$ is the step size, $\xi_j(h)$ is the auxiliary variable and $\zeta(h)$ the compact form, (23a) is the step of solving Agg's or EU's subproblem, and (23b)–(23c) are Agg's steps of updating Lagrangian multiplier.

C. Entire Algorithm

The primitive HDEM algorithm is shown in Algorithm 1. Most distributed optimization algorithms including Algorithm 1 require a perfect operational condition. However, unexpected incidents usually happen in practical systems. Next, we will improve the algorithm to tackle imperfect conditions.

Remark 1 (Distributed Implementation): The DSO is not necessary for Algorithm 1, which can be implemented in a fully distributed manner. In Step S2, the computational burden originates from the multiplication items $A\mathbf{p}(k)$ and $A^T\lambda(k+1)$, where matrix A is derived from the LinDistFlow model. Reference [28] proposed a distributed algorithm for the multiplication in the LinDistFlow model. Embedded by the algorithm, DESs can update the dual and auxiliary variables without the DSO. Each DES only needs to communicate with its neighbors. Meanwhile, the computational burden is distributed to the DESs.

IV. AN ERROR-TOLERANT AND ASYNCHRONOUS HDEM

In this paper, an ETA-HDEM algorithm is proposed to address two imperfect issues during iterations: communication asynchrony and solution inexactness.

In the HDEM architecture in Fig. 1, each agent (including DSO, DG, DES, Agg, and EU) is equipped with a smart processor to make decisions and transmit information through

Algorithm 1 Primitive HDEM Algorithm

Input: Error tolerances $\epsilon_1, \epsilon_2 > 0$ and initial values of dual variables $\lambda(0), \xi(0)$.

Output: Optimal power injected of DERs $\mathbf{p}^*$ and load profile of EUs $\mathbf{p}^{eu*}$.

For $k = 0, 1, \dots$

S1: DER i receives $\mu_i(k)$, calculates the power output profile $\mathbf{p}_i(k)$, and sends it to DSO. Specifically, DGs and DESs can directly solve (20a), while Aggs implement the following inner-loop iteration:

for $h = 0, 1, \dots$

S1.1: Agg or EU $j \in \mathcal{U}_i^+$ receives $\xi_j(h)$, solves its subproblem to obtain a new power profile $\mathbf{p}_j^a(h)$ by (23a), and uploads it to the Agg.

S1.2: Agg i updates $\xi(h+1)$ by (23b), calculates the auxiliary variable $\zeta(h+1)$ and broadcasts it to EUs in $\mathcal{U}_i$.

S1.3: If $\|\xi(h+1) - \xi(h)\| \leq \epsilon_2$, terminate the inner iteration and set $\mathbf{p}_i(k) \leftarrow \mathbf{p}_i^a(h)$ and $\mathbf{p}_j^{eu}(k) \leftarrow \mathbf{p}_j^a(h)$.

end for

S2: DSO updates the dual variable $\lambda(k+1)$ by (20b), calculates the auxiliary variable $\mu(k+1)$ by (20c) and then broadcasts it to DERs.

S3: If $\|\mu(k+1) - \mu(k)\| \leq \epsilon_1$, terminate the outer iteration and set $\mathbf{p}_i^* \leftarrow \mathbf{p}_i(k)$ and $\mathbf{p}_j^{eu*} \leftarrow \mathbf{p}_j^{eu}(k)$.

end for

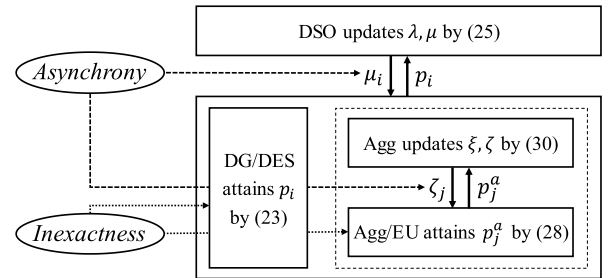


Fig. 2. Information flow and operations of agents in the HDEM architecture.

the information flow. Meanwhile, a memory cell is installed to store the latest received information. During iterations, an agent can implement its procedure with the previously stored information if the current is not received. It means that each agent can implement its process according to its own clock. The information flow and operations of agents is shown in Fig. 2.

Communication Asynchrony: In distributed iterative algorithms, it is quite difficult to maintain a synchronous iteration procedure due to time delay, packet loss, and different computational speeds. The asynchronous iterative strategy is faster and more robust than that of waiting for the slowest information [29, Ch. 7]. In the asynchronous iterative framework, the times when agents implement their steps are inconsistent. Let $\mathcal{K}_i \subseteq \{0, 1, \dots\}$ denote the set of iterative indexes when DER i solves the subproblem with the previously stored Lagrangian multipliers. If $k \notin \mathcal{K}_i$, Agg i keeps its solution unchanged. Similarly, let $\mathcal{K}_D \subseteq \{0, 1, \dots\}$ denote

the set of iterative indexes when DSO updates Lagrangian multipliers. If $k \notin \mathcal{K}_D$, DSO keeps Lagrangian multipliers unchanged. Similarly, in the inner-loop iteration, denote by $\mathcal{H}_j$ and $\mathcal{H}_D$ the sets of iterative indexes when (23a) and (23b)–(23c) are implemented, respectively.

Solution Inexactness: When solving subproblems, there will be inevitably solution errors due to the preset error tolerances of the solver or the accuracy of parameters. Moreover, the inner-loop iteration can also introduce errors to the outer one. Solution inexactness may cause a remarkable error or even divergence. To combat solution inexactness, the averaged sequence of inexact solutions during iterations is applied in practice. Solution inexactness is quantified by the definition of ε -suboptimality [30, Sec. 4.1.1].

Definition 1 (ε -Suboptimal): Given $\varepsilon \leq 0$, an inexact solution $\tilde{\mathbf{x}} \in \mathcal{X}$ to the feasible optimization problem $\min_{\mathbf{x} \in \mathcal{X}} f(\mathbf{x})$ is ε -suboptimal if

$$f(\tilde{\mathbf{x}}) - f(\mathbf{x}^*) \leq \varepsilon,$$

where $\mathbf{x}^*$ is an optimal solution.

A. Outer-Loop Iteration

If $k \in \mathcal{K}_i$, DER i will solve the following subproblem

$$\min_{\mathbf{p}_i \in \mathcal{P}_i} f_i(\mathbf{p}_i) + \hat{\boldsymbol{\mu}}_i^T(k) \mathbf{p}_i \quad (24)$$

to attain a solution $\tilde{\mathbf{p}}_i(k)$, which may be not optimal. Following [22], [31], [32], we make the following assumption on the solution inexactness.

Assumption 2: The solution to the subproblem (24) of DER i is ε_i -suboptimal, $\forall i \in \mathcal{N}$.

The previously stored auxiliary variable $\hat{\boldsymbol{\mu}}_i(k)$ is defined as

$$\hat{\boldsymbol{\mu}}_i(k) = \boldsymbol{\mu}_i(k - \delta_{i,k}^d), \quad (25)$$

where $\delta_{i,k}^d$ quantifies the asynchrony level. If $k \notin \mathcal{K}_i$, keep the solution unchanged. DGs and DESs can directly solve (24), while Aggs have to implement the inner-loop iteration with EUs.

Parameter ε_i is used to quantify the level of solution inexactness. Define by $\varepsilon_D := \sum_{i \in \mathcal{N}} \varepsilon_i$ the sum of the maximal inexactness level of DERs.

If $k \in \mathcal{K}_D$, DSO will update Lagrangian multipliers and the auxiliary variable as

$$\boldsymbol{\lambda}(k+1) = [\boldsymbol{\lambda}(k) + \alpha(A\hat{\mathbf{p}}(k) - \mathbf{b})]^+ \quad (26a)$$

$$\boldsymbol{\mu}(k+1) = A^T \boldsymbol{\lambda}(k+1). \quad (26b)$$

The previously stored injected power $\hat{\mathbf{p}}(k)$ is defined as

$$\hat{\mathbf{p}}(k) = (\hat{\mathbf{p}}_1^T(k), \hat{\mathbf{p}}_2^T(k), \dots, \hat{\mathbf{p}}_N^T(k))^T \quad (27a)$$

$$\hat{\mathbf{p}}_i(k) = \tilde{\mathbf{p}}_i(k - \delta_{i,k}^p), \quad i = 1, 2, \dots, N, \quad (27b)$$

where $\delta_{i,k}^p$ quantifies the asynchrony level. If $k \notin \mathcal{K}_D$, keep all variables unchanged.

Following [2], [4], [33], we assume that there exists a maximal interval between the previously stored and the latest information at iterative steps of DSO and DERs.

Assumption 3: In the asynchronous version of the outer-loop iteration (24)–(27), there exist $k_0 \geq 0$ such that

$$0 \leq \delta_{i,k}^d \leq k_0, \quad \forall i \in \mathcal{N}, k = 1, 2, \dots \quad (28a)$$

$$0 \leq \delta_{i,k}^p \leq k_0, \quad \forall i \in \mathcal{N}, k = 1, 2, \dots \quad (28b)$$

B. Inner-Loop Iteration

If $h \in \mathcal{H}_j$, Agg or EU j will solve the following subproblem

$$\min_{\mathbf{p}_j^a \in \mathcal{P}_j^a} g_j(\mathbf{p}_j^{eu}) + \hat{\boldsymbol{\xi}}_j^T(h) \mathbf{p}_j^{eu} \quad (29)$$

to attain a solution $\tilde{\mathbf{p}}_j^a(h)$, which may be not optimal.

The previously stored Lagrangian multiplier $\hat{\boldsymbol{\xi}}_j(h)$ is defined as

$$\hat{\boldsymbol{\xi}}_j(h) = \boldsymbol{\xi}_j(h - \delta_{j,h}^d), \quad (30)$$

where $\delta_{j,h}^d$ quantifies the asynchrony level. If $h \notin \mathcal{H}_j$, keep the solution unchanged.

If $h \in \mathcal{H}_D$, Agg i will update the Lagrangian multiplier as

$$\boldsymbol{\xi}(h+1) = [\boldsymbol{\xi}(h) + \alpha^a(C\hat{\mathbf{p}}^a(h) - \mathbf{d})]^+ \quad (31a)$$

$$\boldsymbol{\zeta}(h+1) = C^T \boldsymbol{\xi}(h+1). \quad (31b)$$

The previously stored value $\hat{\mathbf{p}}_j^a(h)$ is defined as

$$\hat{\mathbf{p}}_j^a(h) = \tilde{\mathbf{p}}_j^a(h - \delta_{j,h}^p), \quad (32)$$

where $\delta_{j,h}^p$ quantifies the asynchrony level. If $h \notin \mathcal{H}_D$, keep the Lagrangian multiplier unchanged.

Similarly to the outer-loop iteration, we make the following assumptions on the inner loop.

- Assumption 4:* 1) The operational cost $g_j(\mathbf{p}_j^{eu})$ of EU j is c_j -strongly convex and differentiable, $\forall j \in \mathcal{U}^+, i \in \mathcal{N}$.
2) The feasible region $\mathcal{P}_j^a$ of EU j is nonempty, compact, and convex, $\forall j \in \mathcal{U}^+, i \in \mathcal{N}$.
3) There exists a strictly feasible interior point in (21), $\forall i \in \mathcal{N}$.

Assumption 5: The solution to the subproblem (29) of DER i is ε_j -suboptimal, $\forall j \in \mathcal{U}^+, i \in \mathcal{N}$.

Assumption 6: In the asynchronous version of the inner-loop iteration (29)–(32), there exist $h_0 \geq 0$ such that

$$0 \leq \delta_{j,h}^d \leq h_0, \quad \forall j \in \mathcal{U}^+, i \in \mathcal{N}, h = 1, 2, \dots, \quad (33a)$$

$$0 \leq \delta_{j,h}^p \leq h_0, \quad \forall j \in \mathcal{U}^+, i \in \mathcal{N}, h = 1, 2, \dots, \quad (33b)$$

C. Entire Algorithm

To tackle solution inexactness, averaging is leveraged in the error-tolerant algorithm. For instance, in the outer-loop iteration, instead of the inexact solution sequence $\{\tilde{\mathbf{p}}(k)\}$, the running average sequence $\{\bar{\mathbf{p}}(k)\}$ is applied for the result, which is defined as

$$\bar{\mathbf{p}}(k) = \frac{1}{|\mathcal{K}_D(k)|} \sum_{k' \in \mathcal{K}_D(k)} \tilde{\mathbf{p}}(k') \quad (34)$$

with $\mathcal{K}_D(k) := \{\kappa \leq k \mid \kappa \in \mathcal{K}_D\}$. Similarly, averaging sequences are embedded in the inner-loop iteration with $\mathcal{H}_D(h) := \{n \leq h \mid n \in \mathcal{H}_D\}$.

The ETA-HDEM algorithm against communication asynchrony and solution inexactness is shown in Algorithm 2.

Algorithm 2 ETA-HDEM Algorithm Against Asynchrony and Inexactness

Input: Error tolerances $\epsilon_1, \epsilon_2 > 0$ and initial values of dual variables $\lambda(0), \xi(0)$.

Output: Near-optimal power injected of DERs $\bar{p}$ and load profile of EUs $\bar{p}^{eu}$.

For $k = 0, 1, \dots$

S1: If $k \in \mathcal{K}_i$, DER i attains a (maybe inexact) solution $\tilde{p}_i(k)$ to (24) and send it to DSO; otherwise, keep it unchanged. DGs and DESs can directly solve subproblems, while Aggs implement the following inner-loop iteration:

for $h = 0, 1, \dots$

S1.1: If $h \in \mathcal{H}_j$, Agg or EU $j \in \mathcal{U}_i^+$ attains a (maybe inexact) solution $\tilde{p}_j^a(h)$ to (29) and send it to the Agg; otherwise, keep it unchanged.

S1.2: If $h \in \mathcal{H}_D$, Agg i implements (31) and broadcasts $\xi_j(h+1)$ to EUs in $\mathcal{U}_i$; otherwise, keep it unchanged.

S1.3: If $h \in \mathcal{H}_D$ and $\|\xi(h+1) - \xi(h)\| \leq \epsilon_2$, terminate the inner iteration and set

$$\tilde{p}_i(k) \leftarrow \frac{1}{|\mathcal{H}_D(h)|} \sum_{h' \in \mathcal{H}_D(h)} \tilde{p}_i^a(h') \quad (35a)$$

$$\tilde{p}_j^{eu}(k) \leftarrow \frac{1}{|\mathcal{H}_D(h)|} \sum_{h' \in \mathcal{H}_D(h)} \tilde{p}_j^a(h'). \quad (35b)$$

end for

S2: If $k \in \mathcal{K}_D$, DSO implements (26) and broadcasts $\mu_i(k+1)$ to DER i ; otherwise, keep Lagrangian multipliers and auxiliary variables unchanged.

S3: If $k \in \mathcal{K}_D$ and $\|\mu(k+1) - \mu(k)\| \leq \epsilon_1$, terminate the outer iteration and set

$$\bar{p} \leftarrow \frac{1}{|\mathcal{K}_D(k)|} \sum_{k' \in \mathcal{K}_D(k)} \tilde{p}(k') \quad (36a)$$

$$\bar{p}^{eu} \leftarrow \frac{1}{|\mathcal{K}_D(k)|} \sum_{k' \in \mathcal{K}_D(k)} \tilde{p}^{eu}(k'). \quad (36b)$$

end for

Due to the asynchronous iterative paradigm, all steps are implemented in parallel.

D. Convergence Analysis

The convergence of the ETA-HDEM algorithm under communication asynchrony and solution inexactness is proved in Theorem 1. Let A_i denote the i th sliced block of $A = (A_1, A_2, \dots, A_N)$. Define $L_D = \sum_{i \in \mathcal{N}} \|A_i\|^2 / c_i$ and $\gamma_\alpha = \frac{1}{\alpha} - (2k_0 + \frac{1}{2})L_D$.

Theorem 1: Suppose Assumptions A1-A6 hold. If the step size satisfies

$$0 < \alpha < \min \left\{ \frac{1}{(2k_0 + 1/2)L_D}, \frac{1}{4L_D} \right\}, \quad (37)$$

Algorithm 2 satisfies:

- 1) *Feasibility:* The violation of constraints diminishes to zero.

- 2) *Near-Optimality:* The objective value converges to a neighborhood of the optimal value.

The proof of Theorem 1 can be found in the Appendix.

Remark 2 (Interaction between Asynchrony and Inexactness): From a qualitative perspective, the communication asynchrony leads to outdated information, while the solution inexactness causes inaccurate information. The interaction between asynchrony and inexactness enlarges the inaccuracy and makes it impossible to distinguish the inaccuracy sources. From a theoretical perspective, the Appendix provides the bounds of the objective function and the violation of constraints with respect to the parameter of communication asynchrony k_0 and the parameter of solution inexactness ϵ_D . The interaction between asynchrony and inexactness results in a slower convergence speed and larger convergence error in the objective value.

Remark 3 (Effect of Cost Parameters): The cost parameters may influence the convergence performance of the ETA-HDEM algorithm against communication asynchrony and solution inexactness. Specifically, the value of strong convexity of the objective function, c_i , plays an important role. On the one hand, as c_i increases, the parameter L_D decreases, indicating a larger range of the step size by (37). Meanwhile, the parameter γ_α increases. On the other hand, decreasing L_D and increasing γ_α means the bounds of the objective function and the violation of constraints are tightened by the Appendix. However, the cost parameters also influence the optimal solution and the corresponding Lagrangian multiplier λ^* , which also affects the convergence. The influence on the Lagrangian multiplier is not determinable. Thus the effect of cost parameters is case-by-case in fact.

E. Algorithm Acceleration

This subsection aims at promoting the convergence performance of Algorithm 2. Theoretically, the dual decomposition algorithm is a gradient ascent method with respect to the dual variable to solve the dual problem [34]. Thus gradient-based acceleration methods can be embedded in Algorithm 2. Here we utilize the well-known Nesterov's acceleration method [35], [36], [37].

In the outer-loop iteration, the step of updating Lagrangian multipliers (26) is replaced by

$$\lambda^{au}(k+1) = [\lambda(k) + \alpha(A\hat{p}(k) - b)]^+ \quad (38a)$$

$$\lambda(k+1) = \lambda^{au}(k+1) + \beta_k(\lambda^{au}(k+1) - \lambda^{au}(k)) \quad (38b)$$

$$\mu(k+1) = A^T \lambda(k+1), \quad (38c)$$

where $\lambda^{au}(k)$ and $\beta_k \geq 0$ are the auxiliary variable and the acceleration rate, respectively.

In the inner-loop iteration, the step of updating Lagrangian multipliers (31) is replaced by

$$\xi^{au}(h+1) = [\xi(h) + \alpha^a(C\hat{p}^a(h) - d)]^+ \quad (39a)$$

$$\xi(h+1) = \xi^{au}(h+1) + \beta_h^a(\xi^{au}(h+1) - \xi^{au}(h)) \quad (39b)$$

$$\zeta(h+1) = C^T \xi(h+1), \quad (39c)$$

where ξ^{au} and β_h^a are the auxiliary valuable and the acceleration rate, respectively.

TABLE I
PARAMETERS OF DESS

Node	$\bar{P}_i^{DES}$ / kW	$\bar{E}_i$ / kWh	$\underline{E}_i$ / kWh	η_i^{dc}, η_i^{dc}
72, 87	30	100	40	0.97
44, 106	60	150	50	0.95

V. CASE STUDY

In this section, simulation experiments are carried out in MATLAB 2021A on a desktop with an Intel i7-12700 CPU and 32 GB memory. Optimization problems are formulated via the toolbox YALMIP and solved by the solver Gurobi.

A. Setup

The IEEE 123-bus system is utilized to validate the effectiveness of Algorithm 2. The topology of the system is shown in Fig. 3. There are 20 DGs, 4 DES, 40 Aggs, and 5440 EUs in the distribution system. The lower and upper bounds of bus voltage are 0.95 p.u. and 1.05 p.u., respectively. In the 123-bus system, there are 40 buses with active load, which are integrated by an Agg. According to the value of the active load at an Agg bus, divide the Aggs into five categories, which serve 80, 120, 160, 200, and 240 EUs, respectively. The topologies of local communities of these five types of Aggs are shown in Fig. 4, where each node is connected to 40 EUs. The capacity of a distribution line in local communities is 240, 420, 660, 840, 1080, or 1260 kW according to the amount of EU it serves. Consider a discrete time interval of 24 hours from 9:00 a.m. to 8:00 a.m. of the next day with $\Delta t = 15$ min. Fig. 5 shows the normalized load profile during this period. Set the step sizes $\alpha = 4000$, $\alpha^a = 0.05$ and the acceleration rates $\beta_k = 0.5$ and $\beta_h^a = 0.5$.

The parameters of DG's operational cost follow the IEEE 14-bus distribution system. The minimal power output of DG is 0, while the maximal is randomly selected in the range of [0.5, 2] MW. The ramping rate is set to [0.5, 0.8] time of the maximal power output. The parameter of DES's operational cost follows [27] with $a_i^{ch} = a_i^{dc} = 1.6 \times 10^{-3} \$/kW^2$. Other parameters of DESs can be found in Table I. The parameter of Agg's operational cost follows [4] with $a_i^{Agg} = 10^{-4} \$/kW^2$. The parameters $P_{i,t}^{Agg} = 0, \forall t$, which means the objective of Agg is flattening the total load profile. The parameter of EU's dis-utility function follows [12] with $a_i^{EU} = 10^{-3} \$/kW^2$. The expected power consumption of an EU is the expected CL plus its default power consumption of DL. For CLs, the expected load is the normalized load profile multiplied by its baseline capacity, which is randomly selected in the range of [2.5, 5] kW. The lower and upper bounds are randomized from 90% to 105% expected load. For DLs, the maximal power and the final energy requirement are uniformly distributed over [2.5, 5] kW and [15, 30] kWh, respectively. The start and termination time of the energy requirement are randomly assigned repeatedly until it is sufficient to accomplish the energy task. The range of start time is [17:00, 22:00], while the termination time is from 17:00 to 8:45 of the next day. The default power consumption of DL is charging by the maximal power from the start time until the final energy requirement is satisfied.

To simulate the asynchronous condition, for the outer-loop iteration, the iteration intervals of two arbitrary adjacent

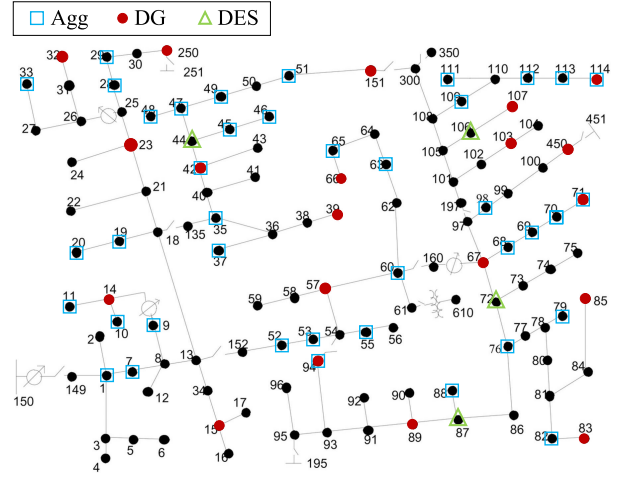


Fig. 3. Configuration of the IEEE 123-bus system.

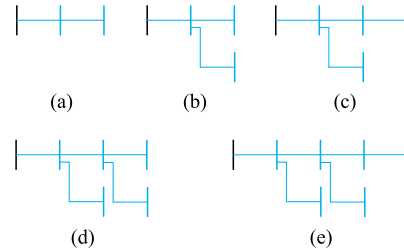


Fig. 4. Topologies of local community systems in different types of Aggs.

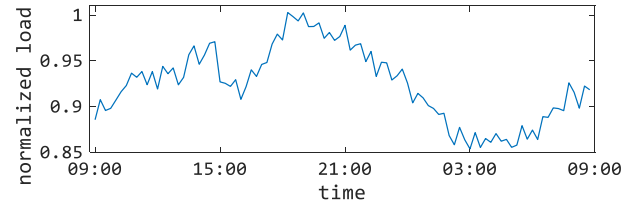


Fig. 5. Normalized load profile.

elements in $\mathcal{K}_i$ or $\mathcal{K}_D$ are uniformly distributed over $[0, k_0]$. Set the asynchrony parameter $k_0 = 10$. The inner-loop iteration is formulated in the same way with $h_0 = 6$. For the solution inexactness, the total inexactness level is set to 3% of the optimal value. To simulate the inexactness condition, every time an agent solves its subproblem, we first attain the optimal solution. Then each element of the optimal solution is multiplied by $(1 + 0.4\tilde{w})$, where $\tilde{w}$ is a standard normal-distribution noise. If the inexact solution does not satisfy Assumptions A2 or A5, halve the noise and repeat the above processes.

B. Main Results

1) *Algorithm Convergence*: To verify the effectiveness and efficiency of the proposed algorithm, we formulate four scenarios:

- *Scenario S0*. Ignore communication asynchrony² and ignore solution inexactness.³

²Always wait for the slowest information.

³Directly use the inexact solutions.

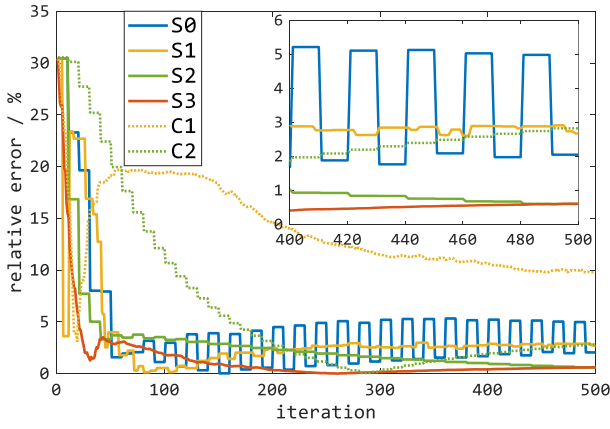


Fig. 6. Relative error of objective value during the outer iteration in the 123-bus system.

- *Scenario S1*. Deal with communication asynchrony⁴ but ignore solution inexactness.
- *Scenario S2*. Deal with solution inexactness⁵ but ignore communication asynchrony.
- *Scenario S3*. Simultaneously deal with communication asynchrony and solution inexactness.

The convergence performance of optimality of the whole two-loop algorithm is shown in Fig. 6. Suffering from the inexactness, the objective in S0 is always away from the optimal value. Meanwhile, the curve indicates that there is idle time during iteration, which is a waste of time. In S1, the curve is more smooth than that in S0, since variables are updated more frequently in a “trotting” manner. Compared to the curve in S0, that in S2 is much more robust against solution inexactness with a much smaller relative error after hundreds of iterations. The proposed algorithm in S3 converges fastest and its relative error of objective is the smallest. Use the iterations when the relative error is smaller than 1% as the criterion to judge the convergence speed. The algorithm that deals with communication asynchrony, S3 (140 iterations), converges faster than that ignoring it, S2 (401 iterations). In the outer-loop iteration of S2, the average waiting time of an agent is 43.86% of the overall iterative indexes. In S2, agents always have to implement their steps every 10 iterative index, while the number is 5.58. From the perspective of convergence accuracy, take the average relative error during [400, 500] iterations as the criterion. The algorithms that deal with solution inexactness, S2 (0.76%) and S3 (0.52%), are remarkably better than those ignoring it, S0 (3.50% and oscillation) and S1 (2.81%).

Moreover, the ability to guarantee feasibility also deserves attention, as shown in Fig. 7. Define the constraint violation by $\|[\mathbf{A}\mathbf{p} - \mathbf{b}]^+\|$. The constraint violation in S0 oscillates heavily, which may cause serious operation accidents. The comparison between S1 and S3 shows that the proposed algorithm can better and more stably maintain constraints by applying average variables to combat with solution inexactness. Compared to S2, the constraint violation in S3 converges faster due to the use of the update-with-latest asynchronous framework. To reduce the constraint violation to 0.5 p.u., it takes S2 and S3

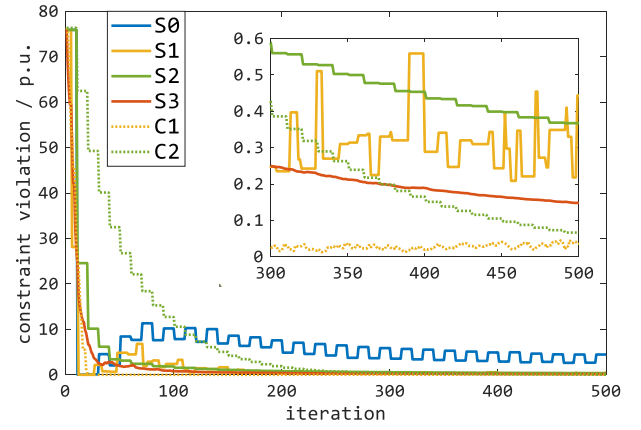


Fig. 7. Constraint violation during the outer iteration in the 123-bus system.

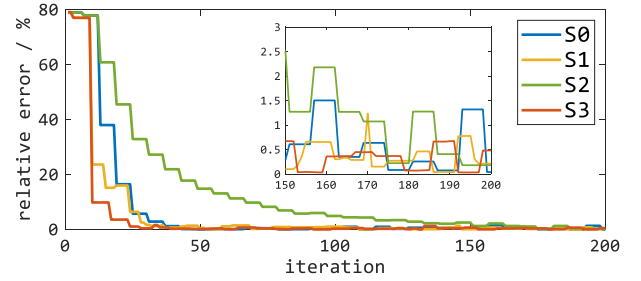


Fig. 8. Objective value during the inner iteration in the 123-bus system.

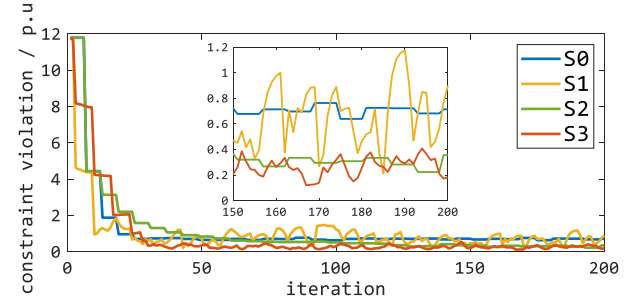


Fig. 9. Constraint violation during the inner iteration in the 123-bus system.

155 and 351 iterations, respectively. Figs. 6 and 7 indicate that the proposed algorithm can attain a feasible and near-optimal solution to the energy management problem in the condition of communication asynchrony and solution inexactness.

Figs. 8 and 9 verify the excellent convergence of the proposed algorithm in the inner-loop iteration combating communication asynchrony and solution inexactness. Consider the Agg at bus 49 with 200 EUs during the 20th iteration of the outer-loop iteration. The constraint violation is defined as $\|[\mathbf{C}\mathbf{p}^a - \mathbf{d}]\|$. For the optimality, despite the curves all converging to a small neighborhood of the optimal value, the proposed algorithm in S3 is obviously the fastest. Moreover, the proposed algorithm is also superior in the effect of maintaining constraints. Take the average constraint violation during [150, 200] iterations as the criterion. The value in S3, 0.26 p.u., is much smaller than those in S0-S2, which are 0.70, 0.67, and 0.30, respectively.

2) *Algorithm Comparison*: To the best of our knowledge, there are no other approaches in the existing literature that can simultaneously cope with communication asynchrony and

⁴Simultaneously update with the latest information.

⁵Average to alleviate solution error.

TABLE II
COST OF ENERGY MANAGEMENT WITH EUS OR NOT

Cost w/o EUs / \$	Cost w/ EUs / \$	Relative Improvement
8.736×10^4	7.563×10^4	15.51%

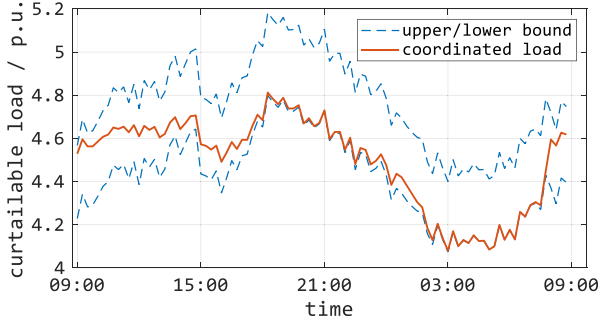


Fig. 10. Total curtailable load profile in the 123-bus system.

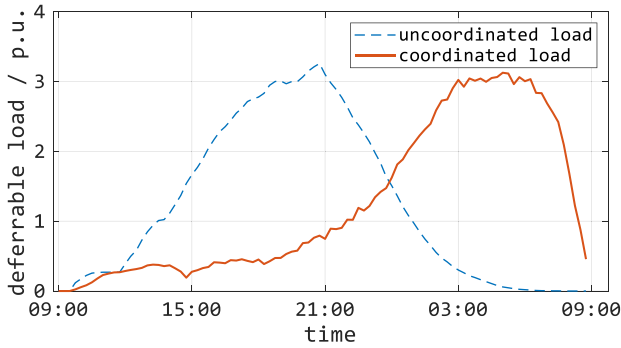


Fig. 11. Total deferrable load profile in the 123-bus system.

solution inexactness. We formulate two other scenarios as comparisons:

- *Scenario C1.* Implement an alternating direction method of multipliers (ADMM) against communication asynchrony proposed by [38]. For solution inexactness, directly use the inexact solutions.
- *Scenario C2.* Implement an augmented Lagrangian method (ALM) against solution inexactness proposed by [32]. For communication asynchrony, always wait for the slowest information.

The step size in Scenario C1 and C2 are both 5. The relative errors of objective values and the constraint violations of C1 and C2 are also shown in Figs. 6 and 7, respectively. After 500 iterations, the relative error of objective value in C1 is 9.82%, still severely deviating from the optimal value, whereas that in S3 is only 0.60%. Given that the constraint violation in C1 is quite small, it converges to a feasible point that suffers from heavy objective loss. Due to the quadratic regularization term in ALM, the violation constraint in C2 is slightly better than that of S3. However, the relative error of C2 after 500 iterations, 2.82%, is almost 4.7 times that of S3.

3) *Effect of EUs:* EUs play an important role in the energy management problem of the distribution system. As can be seen in Table II, with the coordination of EUs, the total operational cost of the energy management problem is reduced by 15.51%. The effect of EUs is inspiring with the rapidly increasing scale.

Figs. 10 and 11 show the total curtailable and deferrable load profiles of EUs, respectively. As can be seen in Fig. 10, the power consumption of curtailable loads is mainly cut down at [18:00, 22:00] and [2:00, 8:00] of the next day, which alleviates the power supply burden. The former is peak time, while the latter is due to a large deferrable load, as shown in Fig. 11. In Fig. 11, the uncoordinated load means that the deferrable load runs at the maximal power from the start time until the energy requirement is satisfied. When EUs participate in energy management, the deferrable load is shifted to valley time, i.e., from 00:00 to 8:00 a.m.. EUs play a significant role in peak cutting and valley filling.

VI. CONCLUSION

In this paper, we study the multi-period energy management problem with massive EUs coordinated by Aggs, forming a hierarchical “DSO-Agg-EU” architecture. To address communication asynchrony and solution inexactness in the energy management problem, we propose an error-tolerant and asynchronous hierarchically distributed algorithm. The proposed algorithm can simultaneously tackle these two imperfect issues, highly surpassing existing literature that can only consider one of them. Besides, we prove the convergence of the algorithm: the total operational cost converges to a near-optimal value, while the violation of physical constraints vanishes. The effect of the proposed algorithm is verified by numerical experiments on the IEEE 123-bus distribution system with 5440 end-users. It shows impressive robustness against communication asynchrony and solution inexactness and excellent scalability in large-scale distribution systems with massive DERs.

It is expected that this study can inspire more scenarios of distribution system operations concerning communication asynchrony and solution inexactness.

APPENDIX

PROOF OF THEOREM 1

We first prove the convergence of the outer-loop iteration. Denote by λ^* a Lagrangian multiplier according to the optimal condition of (17). Let F^* and F^0 denote the sum of all subproblems’ optimal values under λ^* and λ^0 , respectively. In fact, F^* is also the optimal value of the problem (17).

Following from [39, Th. 9], if the step size satisfies (37), we have

$$\begin{aligned} \left\| [A\bar{p}(k) - b]^+ \right\| &\leq \frac{M_{1/2}}{\sqrt{k+1}} + \frac{M_1}{k+1}; \\ -\frac{M_{1/2}\|\lambda^*\|}{\sqrt{k+1}} - \frac{M_1\|\lambda^*\|}{k+1} &\leq \sum_{i \in \mathcal{N}} f_i(\bar{p}_i(k)) - F^* \\ &\leq N_0 + \frac{N_{1/2}}{\sqrt{k+1}} + \frac{N_1}{k+1}; \end{aligned} \quad (40) \quad (41)$$

where $M_{1/2}$, M_1 , N_0 , $N_{1/2}$, and N_1 are constants defined as

$$\begin{aligned} M_{1/2} &:= \frac{2k_0 + 1}{\alpha} \left(2\sqrt{\alpha\epsilon_D} + \frac{2k_0\sqrt{2L_D\epsilon_D}}{\gamma_\alpha} \right) \\ M_1 &:= \frac{2k_0 + 1}{\alpha} \left(2\|\lambda^*\| + \|\lambda^0\| + 2k_0\sqrt{\frac{F^* - F^0}{\gamma_\alpha}} \right) \end{aligned}$$

$$\begin{aligned}
N_0 &:= 2\varepsilon_D + \frac{4k_0^2(2k_0 + 1)L_D\varepsilon_D}{\alpha\gamma_\alpha^2} \\
N_{1/2} &:= \frac{4k_0^2(2k_0 + 1)\sqrt{2L_D\varepsilon_D(F^* - F^0)}}{\alpha\gamma_\alpha^{3/2}} \\
N_1 &:= \frac{2k_0 + 1}{2\alpha} \left(\|\lambda^0\|^2 + \frac{4k_0^2(F^* - F^0)}{\gamma_\alpha} \right).
\end{aligned}$$

Inequalities (40) and (41) guarantee the decision variables generated by the outer-loop iteration are feasible and near-optimal against communication asynchrony and solution inexactness. Since the problem (21) for the inner-loop iteration has the same mathematical structure as the problem (17) for the outer-loop iteration, the convergence of the inner-loop iteration can be directly proved in the same way. Due to the space limitation, the proof is omitted here. The combination of convergences of the outer- and inner-loop iterations completes the proof.

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